SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

BCSE302P - DATABASE SYSTEMS LAB

Semester : Winter Semester 25-26 **Faculty : Dr.S.Geetha**

EXERCISE 7

Title: aViews

1. OBJECTIVE

- To understand the concept of Views in SQL
- To create and use simple and complex views
- To perform data abstraction and security using views
- To apply views for simplified data retrieval
- To understand updateable and non-updateable views

2. VIEWS

- A View is a virtual table based on the result of a SQL SELECT query.
- It does not store data physically; it stores the query definition.
- The data displayed in a view is always up-to-date with the base tables.

In simple terms,

👉 A View is a named SELECT statement stored in the database.

Usage:

- To simplify complex queries
- To provide data security by restricting access to specific columns/rows
- To present customized data to different users
- To maintain consistency in query logic
- To hide complexity of underlying tables

Types:

- A view is created using one or more tables
- Changes in base tables are reflected in the view
- Views can be queried like tables
- Views may or may not be updateable
- ORDER BY is not allowed while creating a view (unless used with LIMIT in some DBs)

3. SYNTAX

```
CREATE VIEW view_name AS
SELECT column1, column2
FROM table_name
WHERE condition;
```

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Using a View

SELECT * FROM view_name;

Dropping a View
DROP VIEW view_name;

CUSTOMER TABLE

customer_id	customer_name	city	phone
501	Ravi	Chennai	9876543210
502	Meena	Coimbatore	9123456780
503	Karthik	Madurai	9988776655
504	Anjali	Trichy	9567891234
505	Suresh	Salem	9898989898

PRODUCT TABLE

product_id	product_name	category	price
1001	Laptop	Electronics	55000
1002	Mobile	Electronics	25000
1003	Chair	Furniture	4500
1004	Table	Furniture	7000
1005	Headphones	Electronics	3000

ORDER TABLE

order_id	customer_id	order_date	total_amount
9001	501	2023-01-10	58000
9002	502	2023-01-15	25000
9003	503	2023-02-05	7000
9004	504	2023-02-18	3000
9005	505	2023-03-01	55000

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ORDER ITEM TABLE

item_id	order_id	product_id	quantity
1	9001	1001	1
2	9001	1005	1
3	9002	1002	1
4	9003	1004	1
5	9004	1005	1
6	9005	1001	1

1. **Create a view to display customer name and city.**

```
CREATE VIEW customer_city_view AS
SELECT customer_name, city
FROM CUSTOMER;
```

2. **VIEW WITH CONDITION**
   - A view can restrict rows using WHERE clause
   - Useful for data security and filtering

A. **Create a view to display electronic products only.**

```
CREATE VIEW electronic_products AS
SELECT product_name, price
FROM PRODUCT
WHERE category = 'Electronics';
```

3. **COMPLEX VIEW (JOIN)**
   - A complex view is created using multiple tables
   - Generally non-updateable

B. **Create a view showing customer name, order date, and total amount.**

```
CREATE VIEW customer_orders AS
SELECT c.customer_name, o.order_date, o.total_amount
FROM CUSTOMER c
JOIN ORDERS o ON c.customer_id = o.customer_id;
```

4. **VIEW WITH AGGREGATE FUNCTION**
   - Views can summarize data using aggregate functions

Create a view showing category-wise average price.

```
CREATE VIEW category_avg_price AS
SELECT category, AVG(price) AS avg_price
FROM PRODUCT
GROUP BY category;
```

E. **UPDATABLE VIEW**

Increase price of all furniture products using a view.

```
UPDATE electronic_products
SET price = price + 500;
```

5. DROPPING A VIEW

```
DROP VIEW customer_city_view;
```

4. EXPECTED OUTCOMES

After completing this exercise, the student will be able to:

- Create and use SQL views
- Apply data abstraction using views
- Restrict data using conditional views
- Work with complex and aggregate views

5. EXPERIMENT QUESTIONS

What is a view?

How does a view differ from a table?

What are the advantages of using views?

When is a view not updateable?

Can we perform INSERT operations on a view?

PRACTICE QUESTIONS

CUSTOMER

customer_id	customer_name	city	phone
101	Ramesh	Chennai	9876543210

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customer_id	customer_name	city	phone
102	Sita	Madurai	9123456789
103	Arjun	Coimbatore	9988776655
104	Kavya	Trichy	9567891234
105	Mahesh	Salem	9898989898

ACCOUNT

account_no	customer_id	branch_name	account_type	balance
5001	101	Chennai Main	Savings	150000
5002	102	Madurai	Current	75000
5003	103	Coimbatore	Savings	200000
5004	104	Trichy	Savings	45000
5005	105	Salem	Current	120000

LOAN

loan_id	customer_id	loan_type	loan_amount	branch_name
9001	101	Home	2500000	Chennai Main
9002	103	Car	600000	Coimbatore
9003	105	Personal	300000	Salem

SIMPLE VIEW QUESTIONS (10)

1. Create a view to display **customer name and city** only.
2. Create a view showing **account number, account type, and balance**.
3. Create a view to display **customers belonging to Chennai city**.
4. Create a view showing **savings account holders** only.

5. Create a view to display **customers with account balance above ₹1,00,000**.
6. Create a view showing **loan id, loan type, and loan amount**.
7. Create a view to display **customers who have taken any loan**.
8. Create a view showing **branch names from the account table without duplicates**.
9. Create a view to display **customers whose phone numbers are not NULL**.
10. Create a view showing **accounts opened in a specific branch**.

B. COMPLEX VIEW QUESTIONS (10)

1. Create a view to display **customer name, account number, and balance** by joining CUSTOMER and ACCOUNT tables.
2. Create a view showing **customer name, branch name, and loan amount** for customers who have taken loans.
3. Create a view to display **branch-wise total account balance**.
4. Create a view showing **customer-wise total loan amount**.
5. Create a view to display **customers who have both an account and a loan**.
6. Create a view showing **branch-wise number of accounts**.
7. Create a view to display **customers having more than one account**.
8. Create a view showing **average account balance for each branch**.
9. Create a view to display **customers whose loan amount exceeds their account balance**.
10. Create a view showing **customers who do not have any loan**.